

Supplementary Material: Regime-Conditioned Dynamical-EVT Diagnostics for Predictive Maintenance

Diogo Ribeiro

July 27, 2026

Contents

1	Extended Methods	1
2	Additional Simulations	2
2.1	Reference availability	2
2.2	Sensitivity to perturbations	2
3	Additional Real-Data Results	4
4	Robustness and Negative Results	5
4.1	Full Threshold and Alarm-Conversion Grids	5
4.2	Full Score-Stratification Bins	7
4.3	Ablation Configuration Registry	7
4.4	Baseline Runner Smoke Audit	7
5	Reproducibility	9

1 Extended Methods

The implemented state-space construction requires causal availability. If a delay embedding is used, every lagged coordinate is shifted so that Z_t contains only information observed at or before t . Scaling parameters, dangerous-region prototypes, and thresholds are fitted on the declared training or calibration partitions and reused without refitting on the test partition.

Runs declustering uses a non-negative integer run length m . If exceedance times are $e_1 < \dots < e_k$, then e_i and e_{i+1} belong to the same cluster when $e_{i+1} - e_i \leq m + 1$. Cluster severity may be summarized by size, duration, maximum observable, or integrated excess above threshold. These empirical quantities should not be equated with fault duration unless the sampling frequency, gap rule, and label definition justify that interpretation.

The inter-exceedance estimator is included because cluster inference can be sensitive to manual declustering choices [2]. The dynamical interpretation follows the hitting-time connection between returns to shrinking target sets and extreme-value laws [3]. The periodicity result motivates the caution that recurrence can reduce the extremal index without implying a specific mechanical cause [4].

Horizon-risk calibration is computed after freezing the event target. The current generated reliability diagram is a diagnostic based on ranked risk surrogates; a full probabilistic calibration claim would require held-out calibration data, uncertainty bands, and explicit acceptance criteria.

2 Additional Simulations

The simulation registry includes independent, periodic, noisy, mixture, missingness, and degradation families. Independent simulations check whether estimators approach the non-clustered target. Periodic simulations check whether the workflow detects clustered returns. Noisy simulations test whether finite perturbations mask recurrence, following the warning that observed noise can move finite samples toward apparent independence [1]. Regime mixtures and preprocessing perturbations test whether clustering can be generated by analysis choices rather than physical degradation.

The generated simulation asset used in the main paper is the reproducibility-scale artifact. It is sufficient for build validation, but it is not a replacement for the full registered experiment matrix. The supplement therefore treats simulation results as evidence about the implemented pipeline and estimator sensitivity, not as an exhaustive finite-sample theory.

2.1 Reference availability

Table 1 records, for every simulated family, how many configurations carry a reference extremal index, whether that reference is theoretical or numerical, and which clustering regime it falls in. Three families — noisy periodic dynamics, persistent shift, and regime mixture — have no established reference at any configuration: the two long-run estimates used to fix a numerical value disagree beyond the accepted tolerance, with discrepancies from 0.15 to 0.60 and no plateau in the block-based family. These are mixture and non-stationary processes for which a single extremal index is arguably not well defined, and they are reported as non-identifiable rather than assigned a value that one declustering convention happens to produce.

The complete estimator grid, carrying all thirteen reported quantities for every configuration and estimator, is distributed as `estimator_full_grid.csv` in the reproducibility package. Every value in the main-paper table is recomputable from it.

2.2 Sensitivity to perturbations

Figure 1 reports mean estimates under noise, missingness, and regime mixture. Added noise moves every estimator towards independence, which is the expected direction: perturbation breaks the deterministic recurrence that produces clustering, and the numerically established references for the noisy families sit between 0.94 and 0.98 accordingly. This is also why the high-replication grid, which runs at a single non-zero noise level, is reported in the main paper as evidence about near-independent processes.

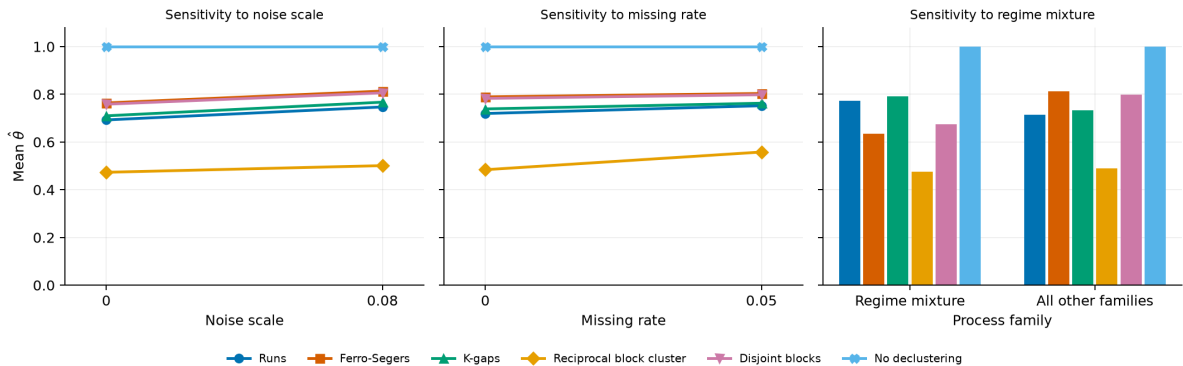


Figure 1: Estimator sensitivity to noise scale, missing rate, and regime mixture. The regime-mixture panel contrasts families rather than sweeping a parameter, since mixture is a property of the process and not a tuning knob.

Table 1: Reference availability and clustering regime for every simulated process family. Configurations without an established reference contribute no bias, RMSE or coverage evidence.

process	configurations	with reference	reference kind	reference theta	regime
burst	1	1	theoretical	1.000	near_independent
missingness					
cyclic	37	24	numerical	0.049, 0.062,	strongly_clustered
degradation				0.086, 0.175	
downsampling	1	1	numerical	0.981	near_independent
iid bounded	37	37	theoretical	1.000	near_independent
tail					
iid light tail	38	38	theoretical	1.000	near_independent
iid pareto	37	37	theoretical	1.000	near_independent
lagged	37	12	numerical	1.000	near_independent
multivariate					
logistic	37	31	numerical,	0.983, 0.984,	near_independent
nonperiodic			theoretical	0.992, 1.000	
target					
logistic	38	31	numerical,	0.500, 0.939,	moderately_clustered,
periodic			theoretical	0.940, 0.970	near_independent
target					
missing at	1	1	theoretical	1.000	near_independent
random					
noisy periodic	1	0	none		not
dynamics			established		established
persistent	1	0	none		not
shift			established		established
regime	37	0	none		not
mixture			established		established
smoothing	1	1	numerical	0.272	strongly_clustered

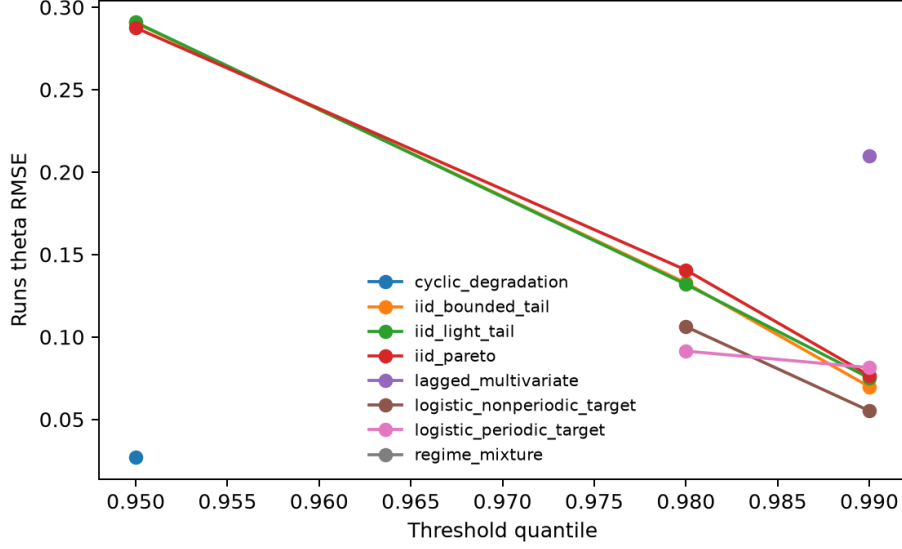


Figure 2: Supplementary copy of the generated simulation RMSE artifact.

3 Additional Real-Data Results

MetroPT, MetroPT2, SCANIA Component X, Hydraulic Systems, and SECOM differ in sampling, entity structure, labels, and exposure. The source registry records acquisition status and provenance for each dataset. MetroPT and MetroPT2 are compressor telemetry resources, SCANIA Component X is a fleet-level truck dataset with anonymized component histories, Hydraulic Systems is an experimental test-rig cycle dataset, and SECOM is a semiconductor process/yield dataset [5–9]. Because the datasets support different estimands and independent units, the industrial evidence is presented as a limitation and reproducibility target rather than a completed population-level comparison.

The event timeline artifact is retained because it can reveal aligned-axis mistakes, alarm bursts outside failure windows, and threshold behavior that aggregate tables may hide. It does not replace event matching, calibration, or independent-unit uncertainty.

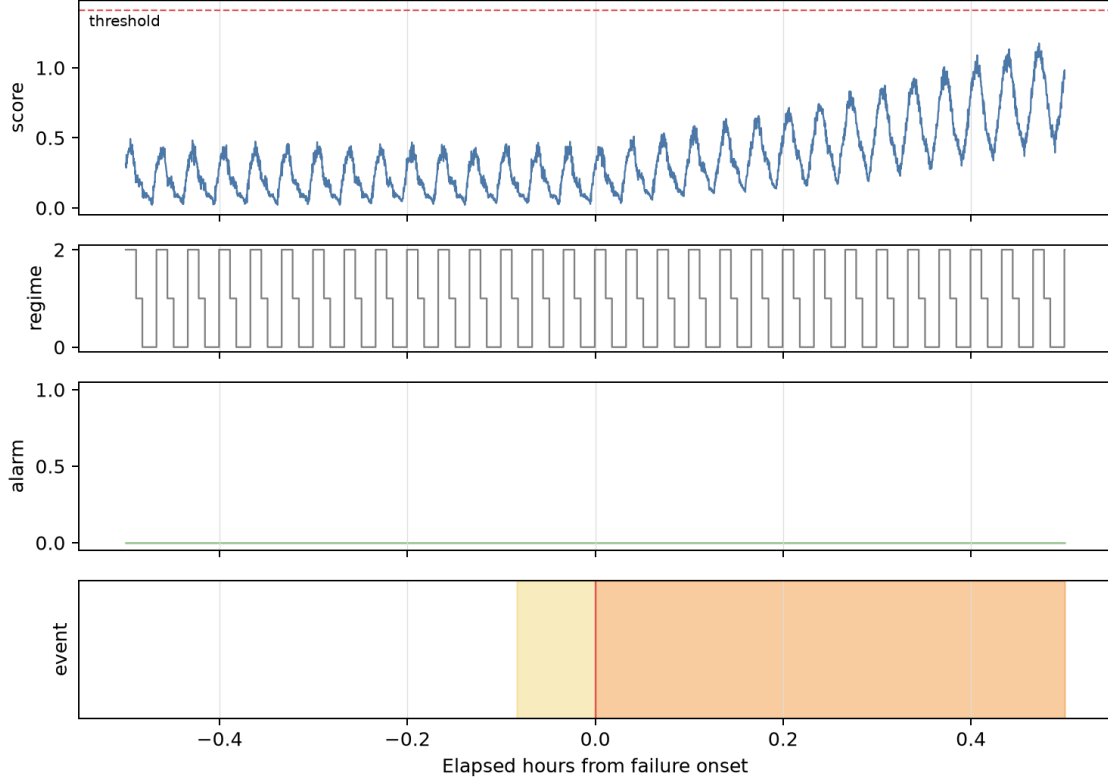


Figure 3: Supplementary alarm timeline generated from the cached processed data.

4 Robustness and Negative Results

The robustness policy is to preserve unfavorable outcomes. Threshold sensitivity, run-length sensitivity, regime-definition changes, dangerous-region definitions, missingness perturbations, and noise perturbations are all represented as registered families in the generated ablation table. Some cells are summaries of planned or non-estimable effects rather than completed industrial contrasts, so the ablation table is treated as a coverage record rather than a positive performance result.

Negative and null findings are part of the evidence. The manuscript does not turn diagnostic evidence into broad industrial performance claims, and it does not use empirical reliability diagnostics as calibrated probability evidence. These constraints are enforced by the paper-check command and by generated audit artifacts.

4.1 Full Threshold and Alarm-Conversion Grids

Table 2: Failed-detection grid for threshold, recall, precision, and false-alarm burden.

threshold quantile	threshold	event recall	event precision	false alarm events per day	median warning lead time
0.9	1.119	0	0	168.5	
0.95	1.329	0	0	146.9	
0.97	1.388	0	0	133.9	
0.98	1.411	0	0	146.9	
0.99	1.439	0	0	138.2	

Table 3: Threshold and run-length stability diagnostics.

threshold quantile	run length	n exceedances	n clusters
0.9	0	2000	90
0.9	2	2000	41
0.9	5	2000	39
0.9	10	2000	38
0.9	20	2000	38
0.95	0	1000	105
0.95	2	1000	43
0.95	5	1000	34
0.95	10	1000	33
0.95	20	1000	33
0.97	0	600	141
0.97	2	600	44
0.97	5	600	31
0.97	10	600	31
0.97	20	600	31
0.98	0	400	157
0.98	2	400	48
0.98	5	400	34
0.98	10	400	31
0.98	20	400	31
0.99	0	200	128
0.99	2	200	68
0.99	5	200	32
0.99	10	200	31

Table 4: Full alarm-conversion grid for merge gap and post-onset matching tolerance.

merge	tol after	recall	prec	F1	false alarms	dup	FA/day	lead	warn time
0	0	0	0	0	157	0	678.2		400
0	30	0	0	0	157	0	678.2		400
0	60	0	0	0	157	0	678.2		400
0	120	0	0	0	157	0	678.2		400
0	300	0	0	0	157	0	678.2		400
5	0	0	0	0	34	0	146.9		586
5	30	0	0	0	34	0	146.9		586
5	60	0	0	0	34	0	146.9		586
5	120	0	0	0	34	0	146.9		586
5	300	0	0	0	34	0	146.9		586
20	0	0	0	0	31	0	133.9		604
20	30	0	0	0	31	0	133.9		604
20	60	0	0	0	31	0	133.9		604
20	120	0	0	0	31	0	133.9		604
20	300	0	0	0	31	0	133.9		604
60	0	0	0	0	31	0	133.9		604
60	30	0	0	0	31	0	133.9		604
60	60	0	0	0	31	0	133.9		604
60	120	0	0	0	31	0	133.9		604
60	300	0	0	0	31	0	133.9		604

4.2 Full Score-Stratification Bins

Table 5: Split-aware score-stratification intervals using training-ranked scores.

split	lower	upper	count	pos	mean rank	obs prop	CI low	CI high
test	0.9	1	4000	401	1	0.1003	0.09132	0.1099
train	0	0.1	1199	0	0.05	0	0	0.003194
train	0.1	0.2	1200	0	0.15	0	0	0.003191
train	0.2	0.3	1201	0	0.25	0	0	0.003188
train	0.3	0.4	1199	0	0.35	0	0	0.003194
train	0.4	0.5	1200	0	0.45	0	0	0.003191
train	0.5	0.6	1201	0	0.55	0	0	0.003188
train	0.6	0.7	1200	0	0.65	0	0	0.003191
train	0.7	0.8	1199	0	0.75	0	0	0.003194
train	0.8	0.9	1200	0	0.85	0	0	0.003191
train	0.9	1	1201	0	0.95	0	0	0.003188
val	0	0.1	215	52	0.04765	0.2419	0.1895	0.3033
val	0.1	0.2	234	47	0.1494	0.2009	0.1545	0.2568
val	0.2	0.3	219	64	0.2529	0.2922	0.236	0.3556
val	0.3	0.4	280	126	0.352	0.45	0.3928	0.5086
val	0.4	0.5	292	86	0.4522	0.2945	0.2452	0.3492
val	0.5	0.6	283	119	0.5512	0.4205	0.3644	0.4787
val	0.6	0.7	366	197	0.651	0.5383	0.487	0.5887
val	0.7	0.8	323	164	0.7488	0.5077	0.4534	0.5618
val	0.8	0.9	317	134	0.8504	0.4227	0.3696	0.4777
val	0.9	1	1471	1311	0.9883	0.8912	0.8743	0.9061

4.3 Ablation Configuration Registry

4.4 Baseline Runner Smoke Audit

Table 6: Ablation matrix summary.

ablation family	setting	summary value
threshold	0.9	0.1
threshold	0.95	0.05
threshold	0.97	0.03
threshold	0.98	0.02
threshold	0.99	0.01
run_length	0	157
run_length	2	48
run_length	5	34
run_length	10	31
run_length	20	31
lag	0	
lag	1	
lag	2	
lag	5	
lag	10	
regime	off	
regime	loaded	
regime	recovery	
dangerous_region	prototype	
dangerous_region	constraint	
dangerous_region	density	
noise_missingness	none	
noise_missingness	noise_0.05	
noise_missingness	missing_0.05	

Table 7: Synthetic smoke results for the shared baseline runner.

Baseline	Validation F1	Runtime (s)	Parameters
Engineering threshold	0	0.00142	12
Global empirical	0.911	0.00279	1
Regime empirical	0.892	0.0111	3
POT/GPD	0.911	1.77	3
Fixed-run declustering	0.904	0.00892	2
K-gaps declustering	0.904	0.00987	2
SPOT-style	0.919	14.8	4
Isolation Forest	0.92	2.63	120
Robust changepoint	0.566	0.0648	12
Linear autoencoder	0.955	0.0259	54
Conformal anomaly	0.319	0.0162	200
Empirical horizon risk	0.919	0.00306	200

5 Reproducibility

The build is organized around reproducible artifacts. The root command `make paper-assets` regenerates figures, tables, result macros, the results synthesis document, and per-file provenance records. The command `make paper` builds the main manuscript after regenerating assets. The command `make supplement` builds this supplement after the same asset step. The command `make paper-check` compiles both documents and audits generated assets, citations, compiled PDFs, and source markers. The command `make submission-package` assembles the reviewed manuscript package, reviewer report, revision matrix, availability statements, protocol, lock file, and artifact manifests without copying raw third-party datasets.

The generated macros provide stable result values used by the manuscript and supplement. Exact build identifiers and file-integrity records are stored in the repository manifests rather than printed in the narrative text.

Machine-readable interpretation and citation audits are written to `paper/generated/` by the paper checker. Manual edits to generated figures or tables are detected through the artifact-manifest audit.

References

- [1] Davide Faranda, Jorge Milhazes Freitas, Valerio Lucarini, Giorgio Turchetti, and Sandro Vaienti. Extreme value statistics for dynamical systems with noise. *Nonlinearity*, 26(9): 2597–2622, 2013. doi: 10.1088/0951-7715/26/9/2597.
- [2] Christopher A. T. Ferro and Johan Segers. Inference for clusters of extreme values. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 65(2):545–556, 2003. doi: 10.1111/1467-9868.00401.
- [3] Ana Cristina Moreira Freitas, Jorge Milhazes Freitas, and Mike Todd. Hitting time statistics and extreme value theory. *Probability Theory and Related Fields*, 147(3–4):675–710, 2010.
- [4] Ana Cristina Moreira Freitas, Jorge Milhazes Freitas, and Mike Todd. The extremal index, hitting time statistics and periodicity. *Advances in Mathematics*, 231(5):2626–2665, 2012.
- [5] Nikolai Helwig, Eliseo Pignanelli, and Andreas Schütze. Condition monitoring of hydraulic systems, 2018.
- [6] Zahra Kharazian, Tony Lindgren, Sindri Magnússon, Olof Steinert, and Oskar Andersson Reyna. Scania component x dataset: A real-world multivariate time series dataset for predictive maintenance. *Scientific Data*, 12(493), 2025. doi: 10.1038/s41597-025-04802-6.
- [7] Michael McCann and Adrian Johnston. Secom, 2008.
- [8] Bruno Veloso, Rita P. Ribeiro, João Gama, and Pedro Mota Pereira. The metropt dataset for predictive maintenance. *Scientific Data*, 9(764), 2022. doi: 10.1038/s41597-022-01877-3.
- [9] Bruno Veloso, Rita P. Ribeiro, Pedro M. Pereira, and João Gama. Metropt2: A benchmark dataset for predictive maintenance, 2023.